

Until the 17th century, scientists were known as “**natural philosophers**.” Today’s philosophers contend with subjects such as ethics, which cannot be tested by the scientific method.



Geology

Investigating the processes that form rocks and shape our planet’s landscape.



Nuclear chemistry

Studying the behavior of unstable atoms that break apart and release powerful radiation.

PHYSICS

With its name meaning “nature” in Greek, physics is the basis of all other sciences. It provides explanations of energy, mass, force, and light without which other sciences would not make sense.



Particle physics

Studying the particles that make up atoms and carry energy and mass throughout the Universe.



Mechanics

Understanding the motion of objects in terms of mass and how energy is transferred between them by forces.



Wave theory

Explaining sound and other natural phenomena using an understanding of the behavior of waves.



Astronomy

Studying objects, such as planets, stars, and galaxies, in space.

Thermodynamics

Studying the way energy flows through the Universe according to a series of unbreakable laws.



Optics

Studying the behavior of beams of light as they reflect off or shine through different substances.



Electromagnetism

Investigating electric currents and magnetic fields, and their uses in electronic devices.



Meteorology

Understanding the conditions that produce weather.



Social sciences

These sciences are not linked directly with the “natural sciences” (eg, biology, chemistry, or physics). Instead, they apply scientific methods to investigate humanity. Examples include:

Anthropology

Studying the human species, especially how societies and cultures from around the world differ from one another.

Archaeology

Studying ancient civilizations from the remains of their homes and cities.

Economics

Developing theories as to how people and companies spend their money.

Geography

Researching the natural landscape and how humans use the land, such as where they build cities.

Psychology

Investigating the way the human mind works using scientific methods.

Applied science

This area of work takes pure scientific knowledge and uses it for practical purposes. Some applied sciences can be described as types of engineering. Examples include:

Biotechnology

Using the knowledge of genetics and biochemistry to make artificial organisms and biological machines.

Computer science

Building microchip processors and writing software instructions to build faster and smarter computers.

Materials science

Developing new materials with properties suited to a particular application.

Telecommunications

Making use of electromagnetism, radiation, and optics to send signals and information over long distances.